

Dynamic memory allocation

C programming · CM10 · 90 minutes

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- ▶ Arithmetic circuits and circuit generators with **MLIR** / **CIRCT**.
- ▶ **Systolic arrays** and small floating-point formats.
- ▶ Previously: Inria / INSA Lyon, Emeraude; PhD at BSC / UPC.

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Today

1. Allocate an array.
2. Find and fix memory errors.
3. Arrays, strings and matrices.
4. User-defined types.

1. Allocate an array

How much space do we need?

Four integers, then one hundred?

A program has room for **four integers**.

The user asks to store **100**. What must change?

```
demos/00-allocation/main.c
```

```
Terminal: make run
```

Does reading a count create more space?

Predict what happens when we enter **100**.

```
int count;
int p[4];
if (scanf("%d", &count) != 1) return EXIT_FAILURE;
if (count <= 0 || count > 4) {
    fputs("Choose 1 to 4: the array has four elements.\n", stderr);
    return EXIT_FAILURE;
}
```

demos/00-allocation/checkpoints/02-capacity.c

Terminal: make run STEP=02-capacity

Choose the capacity during execution

How can we reserve **count integers**?

```
int *p = malloc(count * sizeof(int));  
if (p == NULL) {  
    fputs("Allocation failed\n", stderr);  
    return EXIT_FAILURE;  
}
```

demos/00-allocation/checkpoints/03-missing-free.c

Terminal: make run STEP=03-missing-free

Count elements, request bytes

```
void *malloc(size_t size);    // From <stdlib.h>  
int *p = malloc(100 * sizeof(int));
```

- ▶ `size_t` is unsigned, with at least **16 value bits**.
- ▶ `sizeof` is an **operator** giving a size in bytes.
- ▶ `sizeof(int)` is the size of one `int`, not the pointer.

Sizes depend on the platform. In C, the `void *` result needs no cast.

2. Find and fix memory errors

The program prints a result. Is it correct?

It prints correctly. Is anything wrong?

Run the allocated version with **100**, then inspect it:

```
make run STEP=03-missing-free  
make valgrind STEP=03-missing-free
```

Where was the unreleased block allocated?

Release the block after its last use

1 · Add the release

```
printf("first=%d, last=%d\n",  
      p[0], p[count - 1]);  
  
free(p);
```

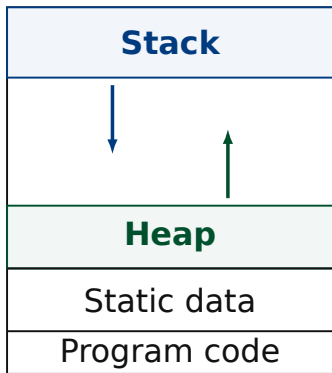
2 · Check again

Rerun Valgrind.

**All heap blocks
freed.**

Where does the program keep its data?

0xffff



count and pointer p
Automatic locals

Room to grow
Opposing directions

p[0] to p[count-1]
Allocated integers

Globals and static

0x0000

Illustrative 16-bit address space; real address widths and layouts vary.

Does malloc fill the array with zeros?

1 · Remove the writes

```
printf("first=%d, last=%d\n",  
      p[0], p[count - 1]);
```

2 · Inspect the read

Uninitialised value

Allocated storage has no initial value.

Write values before reading them

1 · Restore the loop

```
for (int i = 0; i < count; ++i)  
    p[i] = 0;
```

2 · Check again

Rerun Valgrind.

Zero errors

Where does this loop stop?

1 · Predict · count = 4

```
for (int i = 0; i <= count; ++i)  
    p[i] = 0;
```

2 · Inspect

First invalid write:

p[4]

3 · Repair

i < count

What if we free the block too early?

1 · Predict the read

```
free(p);  
printf("first=%d, last=%d\n",  
      p[0], p[count - 1]);
```

2 · Name the problem

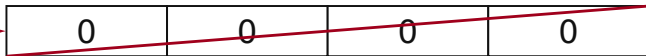
Dangling pointer

The object is gone.

Pointer



Allocated block



Freed storage

Can we still free it after losing its address?

1 · Lose the address

```
p = NULL;  
free(p);
```

2 · Name the problem

Memory leak

The block is still allocated.

Pointer

p

NULL

Allocated block

0	0	0	0
---	---	---	---

Address lost; block still allocated

Local objects can expire too

```
int *bad_result(void) {  
    int value = 42;  
    return &value;  // The local dies when this call ends.  
}
```

A dangling pointer need not come from free

Returning an address does not extend an automatic object's lifetime.

Who controls an object's lifetime?

Static storage

Globals and static objects exist for the **whole program**. Storage is reserved before execution.

Automatic storage

An ordinary local exists during its **block execution**. Each recursive call gets its own locals.

The implementation manages their storage. Deep recursion can **exhaust the stack**.

`demos/14-memory-layout/main.c`
Terminal: `make run`

Who is responsible for reclamation?

C: explicit responsibility

For every successful allocation, decide who will release it and when.
Repeated leaks consume more memory.

Java: garbage collection

The runtime can reclaim unreachable objects automatically.

Four functions in `<stdlib.h>`

Function	Purpose
<code>malloc(bytes)</code>	Allocate uninitialised storage
<code>calloc(count, size)</code>	Allocate storage with all bits zero
<code>realloc(p, bytes)</code>	Resize an allocation; it may move
<code>free(p)</code>	Release an allocation

The free contract

```
void free(void *pointer);    // Declared in <stdlib.h>
```

- ▶ Pass the allocation's **start address**, once.
- ▶ `free(NULL)` does nothing; `free` returns no value.
- ▶ An invalid or repeated release is **undefined behaviour**.

Checkpoint · questions 1-5

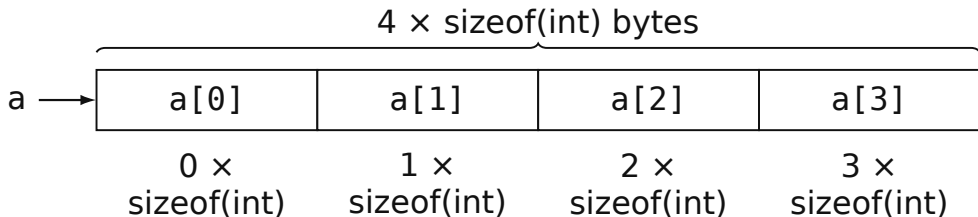
Answer questions **1-5**, then discuss your reasoning.
Open the allocation quiz.

3. Arrays, strings and matrices

Allocate data, then locate each element.

A one-dimensional integer array

```
int *a = malloc(4 * sizeof(int));
```



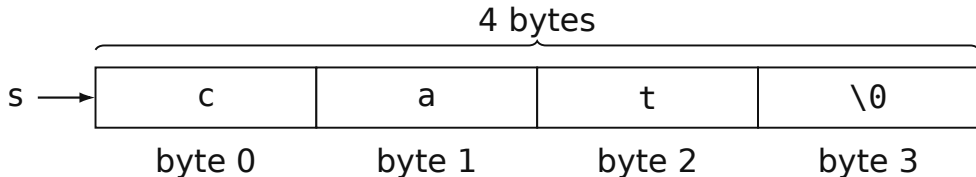
Use `a[i]`, equivalently `*(a + i)`, for **$0 \leq i < 4$** .

`demos/04-array/main.c`

Terminal: `make run`

A character array can hold a string

```
char *s = malloc(4 * sizeof(char));
```



Three letters need **four bytes**, including '`\0`'.

`demos/05-string/main.c`

Terminal: `make run`

Assigning a pointer is not copying text

Predict the values of p and s, and the fate of the allocation.

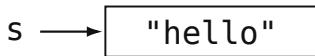
```
char *s = malloc(100);  
if (!s) return EXIT_FAILURE;  
s = "hello";  
char *p = s;  
s = "bye";
```

demos/03-leak/main.c

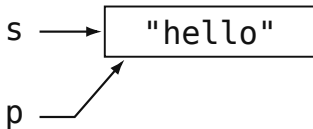
Terminal: make run

Follow the three assignments

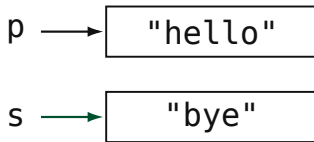
`s = "hello"`



`p = s`



`s = "bye"`



Original heap block: 100 bytes, now unreachable

The literals are separate from the lost heap block. Neither pointer can now be passed to `free`.

Copy characters into owned storage

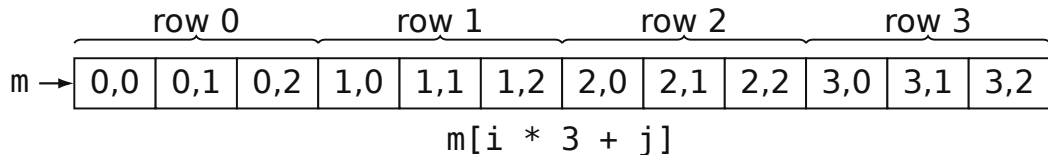
```
char *s = malloc(6);  
if (!s) return EXIT_FAILURE;  
strcpy(s, "hello"); // Include <string.h>.  
printf("%s\n", s);  
free(s);
```

Alternatively, just reference a literal

Use `const char *s = "hello";` when no writable copy is needed.

A 2D array in one block

```
int *m = malloc(4 * 3 * sizeof(int));  
/* After checking m: element (i, j) is m[i * 3 + j]. */
```



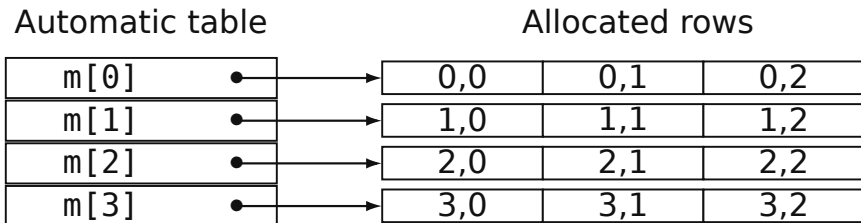
One allocation, one **free(m)**.

demos/06-matrix-flat/main.c

Terminal: make run

An array of row pointers

```
int *m[4]; // Automatic table of four pointers.  
for (int i = 0; i < 4; ++i)  
    m[i] = malloc(3 * sizeof(int));
```



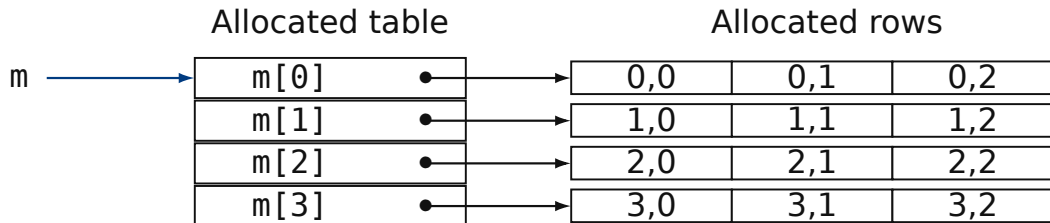
Check each row. Use `m[i][j]`; free each row, **not** `m`.

`demos/07-matrix-rows/main.c`

Terminal: `make run`

Allocate the pointer table too

```
int **m = malloc(4 * sizeof(int *));  
for (int i = 0; i < 4; ++i)  
    m[i] = malloc(3 * sizeof(int));
```



Check `m` first, then each row. Free **rows first**, then `free(m)`.

`demos/08-matrix-dynamic/main.c`

Terminal: `make run`

Checkpoint · questions 6-8

Answer questions **6-8**, then discuss your reasoning.
Open the allocation quiz.

4. User-defined types

Group data and access its members.

An address has a street and a number. How does changing a member affect the object?

Three tools

enum names integer constants. struct groups members. typedef introduces a name for a type.

`demos/19-types/main.c`

Terminal: `make run`

Enumerations

```
enum day { mon, tue, wed, thu, fri, sat, sun };  
enum day today;  
today = wed;
```

By default, values start at **0** and increase by **1**.

Named integer constants

mon is 0; wed is 2; sun is 6.

A structure groups several members

```
struct address {  
    char street[100];  
    int number;  
};
```

A type definition

Members may be integers, floating-point values, arrays, pointers or other structures. No address object exists yet.

Define a type, then create objects

```
struct address home;  
struct address office;
```

These definitions allocate **two distinct objects** of the same type.

Local objects need initialisation

Ordinary local members have indeterminate values until initialised or assigned. The type definition alone sets no values.

Initialising a structure

```
struct address home = { "Paul Bert", 12 };
```

Or combine the type and object definitions:

```
struct address {  
    char street[100];  
    int number;  
} home = { "Paul Bert", 12 };
```

Padding

The implementation may insert unused bytes between members and at the end to satisfy alignment requirements.

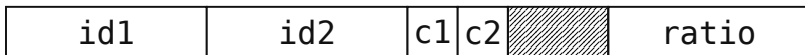
- ▶ Members appear in declaration order.
- ▶ `sizeof(struct T)` includes padding.
- ▶ Layout and alignment depend on the implementation.

Same members, different order

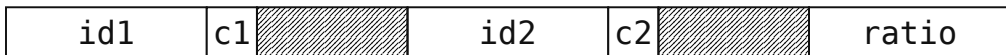
```
struct grouped {  
    int id1, id2;  
    char c1, c2;  
    float ratio;  
};  
struct interleaved {  
    int id1; char c1;  
    int id2; char c2;  
    float ratio;  
};
```

See where the padding goes

Grouped: 16 bytes



Interleaved: 20 bytes



Here: 4-byte int/float, alignment 4; 1-byte char.

16 vs 20 bytes in this layout. Measure with `sizeof` and `offsetof`.

`demos/09-struct-padding/main.c`

Terminal: `make run`

Access members through an object

```
struct address home = { "Paul Bert", 12 };  
strcpy(home.street, "Rue de Paris");  
home.number = 14;
```

The dot operator

Use `.` with a structure object.

`strcpy` needs `<string.h>` and enough space for the terminator.

Access members through a pointer

```
struct address *p = &home;  
strcpy(p->street, "Rue de Paris");  
p->number = 16;
```

The arrow operator

Use `->` with a pointer. `p->number` is the same member as `(*p).number`.

Both forms above modify **home**.

typedef: a name for a type

```
typedef int LENGTH;  
typedef enum { FALSE, TRUE } BOOLEAN;  
typedef struct date DATE;
```

Alias

A typedef introduces a type name, not an object or a distinct underlying type. Then write `LENGTH width;`.

Combine a date and a day of the week

```
struct date { short day, month; int year; };  
typedef enum { mon, tue, wed, thu, fri, sat, sun } DAY;  
typedef struct { DAY day; struct date date; } A_DATE;
```

A nested structure

A_DATE contains a DAY and a complete struct date.

Access nested members

```
A_DATE today = { wed, { 3, 9, 2014 } };  
today.day = wed;  
today.date.year = 2014;  
A_DATE *p = &today;  
p->day = today.day;  
p->date.year = 2026;
```

p->date.year reaches the nested member of **the same object**.

demos/10-nested-struct/main.c

Terminal: make run

Checkpoint · questions 9-12

Answer questions **9-12**, then discuss your reasoning.
Open the allocation quiz.